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Automation of ^{99m}Tc mercaptoacetyltriglycine (MAG3) report writing using a vision language model

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Abstract

Background ^{99m}Tc mercaptoacetyltriglycine (MAG3) studies represent a classic of nuclear medicine. We evaluated the quality of MAG3 reports generated by a vision language model (VLM). The mid-term goal to utilize such systems could be the support of nuclear medicine experts, thereby freeing staff resources currently tied up in administrative and documentation tasks.

Results Human raters generally preferred MAG3 reports written by human authors over those generated using the VLM. Scores for VLM-generated reports improved notably in the subset of cases where the urodynamic curves were correctly identified by the model. Half of the raters found statistically significant differences between human and Qwen-generated reports (Wilcoxon signed-rank test, $\alpha = 0.05$). There was a strong correlation between the assessed text quality and the raters' ability to correctly identify the source of the text (human or Qwen; $R^2 = 0.998$).

Conclusion The VLM-based approach successfully generated MAG3 reports that, to some extent, resembled those written by human experts. However, assessed quality varied among raters. It remains to be seen whether future language model advancements will reach a performance level sufficient for integration into clinical practice.

Introduction

Realistic future scenarios suggest that the demand for diagnostic medical services—whether in nuclear medicine or radiology—will exceed the available imaging capacities [1]. Artificial Intelligence (AI) [2] and vision language models (VLM) could provide valuable solutions by partially automating hospital operations, thereby

freeing healthcare staff capacities. Administrative and documentation tasks associated with medical procedures particularly lend themselves to this technological advancement. Although crucial contextual information, such as patient status, is often present in hospital operations, it is frequently not accessible to software systems. Nonetheless, extensive patient data is already digitally available in hospital information technology (IT) systems, making it suitable for algorithmic processing—with or without AI involvement.

To date, the reliability of language models in nuclear medicine has been limited [3], although research into their application in this field continues. For example, the VLM Pegasus [4] has been tested for tasks such as

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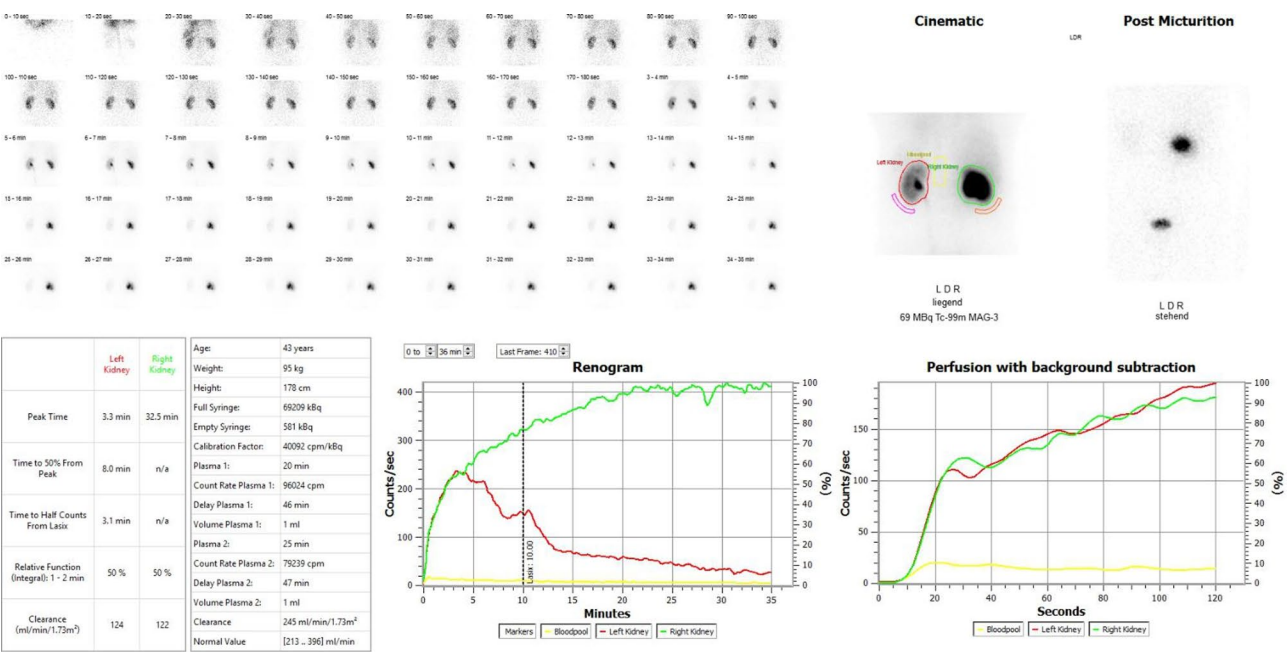


Fig. 1 MAG3 study of case 73 with right kidney showing the accumulation type

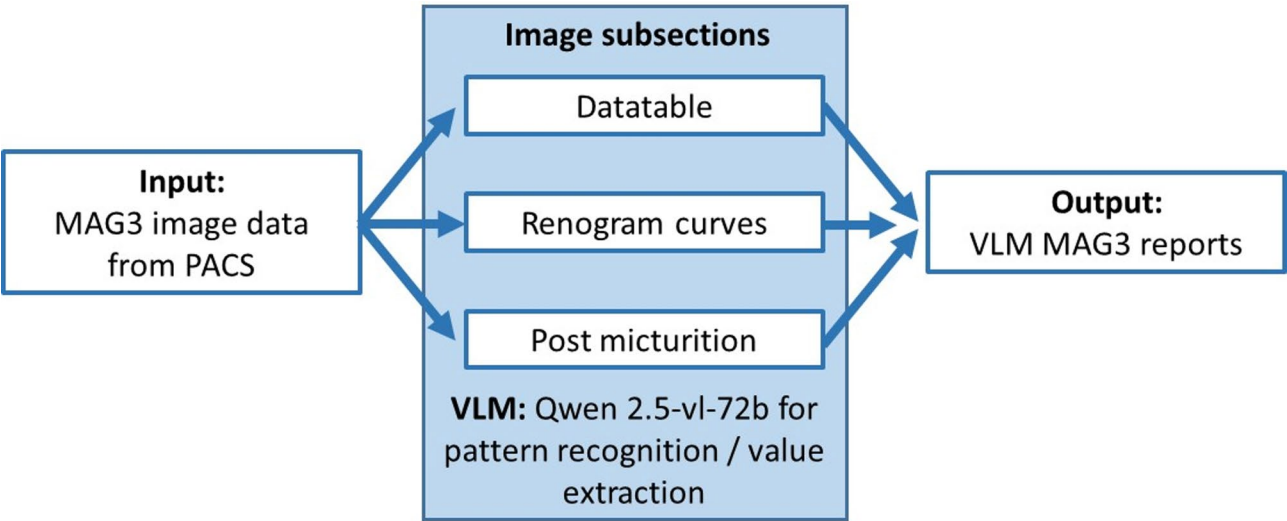


Fig. 2 Flowchart with process broken into manageable sub-tasks

summarizing reports and extracting information from positron emission tomography (PET) studies [5]. Broader research exists in the closely related discipline of radiology. ChatGPT, for instance, has been successfully used to generate simple radiology reports on distal radius fractures [6, 7], and for automated reporting of peripherally inserted central catheter (PICC) line insertions [8, 9]. In the correct diagnosis of e.g. classic musculo-skeletal lesions, its current version is still outperformed by human radiologists [10, 11]. Despite their promise, emerging AI systems exhibit notable limitations, as demonstrated by hallucinations in magnetic resonance

imaging (MRI) reconstruction [12–14], or in the quantification of radiation dose [15].

In this study, we analysed a test set 50 cases of ^{99m}Tc mercaptoacetyltriglycine (MAG3, Fig. 1) studies and evaluated the potential of automated report generation using a state-of-the-art VLM. We selected the Qwen2.5-VL-72b model (provided by Alibaba Cloud, Hangzhou, Beijing, PRC). Qwen 2.5 has previously demonstrated superior performance in medical contexts compared to other models [16]. In this manuscript, Qwen was tasked with analysing the patterns of renal function present in 50 MAG3 studies, please see also the flowchart in Fig. 2.

Human raters then compared the AI-generated reports to the original reports written by nuclear medicine experts retrieved from the Picture Archiving and Communication System (PACS). A preliminary version of this project, utilizing a different VLM, was previously presented as a conference abstract [17–19].

Methods and materials

For this project, a total of 50 MAG3 studies, along with their original PACS reports written by human nuclear medicine experts, were retrospectively obtained. Additionally, prompts were developed for the VLM Qwen. Four raters subsequently participated in a user study to evaluate these reports. The large language model GPT-5 [20] and Claude 4 [21] were used in python code debugging and manuscript optimisation.

MAG3 protocol

All patients were imaged using a Siemens Symbia Intevo Bold dual-head single-photon emission computed tomography (SPECT/CT) system (Siemens Healthineers, Knoxville, TN, USA) operating in single-position planar mode. For ^{99m}Tc imaging, the energy window was centred at 140 keV (width: 128–149 keV). Spatial resolution at the centre of the field of view (FOV) was approximately 8 mm in both transverse and axial planes. Patients were administered a mean activity of 70 MBq (range: 68–74 MBq) of ^{99m}Tc MAG3. The imaging protocol involved the patient lying supine for 40 min and then standing upright for 2 min post-micturition. Images were reconstructed using a matrix of 128×128 pixels across a total of 920 frames. While there is no general standard for timing of Lasix administration [22]; by the internal protocol, Lasix was administered after 10 min.

MAG3 pilot dataset and test dataset

For the present study, a pilot dataset of 95 studies was selected chronologically from MAG3 patients, starting on 01 January 2021. Prompts for the extraction of information and for the generation of reports were developed based on this pilot dataset.

In addition, a total of 50 studies were retrospectively obtained from MAG3 patients for the study's test dataset. Only patients with two kidneys were included. Initially, 20 MAG3 studies were selected starting on 01 January 2022. Due to a high rate of negative findings, an additional 30 MAG3 studies were subsequently included from patients under the condition that at least one kidney exhibited a pathology. All kidneys were explicitly labelled regarding function as either regular, accumulation type (urodynamically relevant), or with decreased function (denoted nephrectomy type). Additionally, kidneys with a non-accumulation type were labelled for delayed peak times. Figure 1 illustrates a sample MAG3 study with

one accumulation-type kidney. The complete image as shown in Fig. 1 was cropped by python code for further processing in Qwen into specific subsections (cinematic, datatable left/right, dynamic, micturition, perfusion, and renogram) to facilitate subsequent processing by Qwen. The automated cropping py-script (Supplement Suppl. 1) applies predefined pixel coordinates to systematically extract distinct regions from each MAG3 image, with the dynamic section capturing the main kidney activity curves and datatable sections isolating the numerical parameter tables for left and right kidneys. This standardized pre-processing ensured consistent input formatting for the VLM while preserving the essential diagnostic information from each functional region of the original PACS image. Explicit written general consent had been obtained from all included patients.

VLM prompting

The current study employed the Qwen2.5-VL-72b model provided by Alibaba Cloud, running locally on four NVIDIA A6000 GPUs (NVIDIA Corporation, Santa Clara, CA, USA). Qwen received PACS image data on a case-by-case basis. The cropped image data of Fig. 1 passed on from the cropping py-script (Suppl. 1) to Qwen contained: the data table for left and for right kidney, the post-void micturition, and the renogram curves. From this image input, Qwen was tasked with classifying kidney function as normal, accumulation type, nephrectomy type, or normal with delayed peak. Furthermore, Qwen had to extract numerical values from the MAG3 data tables such as peak times, relative kidney function, and clearance (Suppl. 2.1 - Suppl. 2.4). A Python script then converted these model outputs into standardized nuclear medicine reports (Suppl. 3 - Suppl. 4). The script applied structured threshold-based decision rules to interpret model-generated values—including peak time, time to 50% activity decay, and Lasix response—and mapped key kidney features (renogram type, elimination pattern, post-void retention, clearance, differential function) to predefined clinical categories using Python enumerations. These categories were arranged into standardized report sections (“Findings,” “Parameters,” “Impression”), composed in consistent, human-readable German-language phrases typically used in clinical routine reporting. Additionally, heuristic merging logic was applied to streamline redundant impressions (e.g., combining symmetric normal function with the absence of obstruction). Thresholds applied by the system defined a normal peak time as 8 min, while the times to a 50% activity decrease and response to Lasix were each defined as 10 min. The overview flowchart is given in supplement Suppl. 5.

User study and python analysis

To assess report quality, a user study was conducted involving four nuclear medicine residents who evaluated the 50 anonymized PACS reports and the 50 anonymized Qwen reports of the test dataset. Raters 1 to 3 were in their final two years of nuclear medicine residency, had completed their scintigraphy rotations, and thus potentially had previous exposure to some of the included MAG3 studies during clinical routine. Rater 4 was in the first half-year of nuclear medicine residency, had not yet completed the scintigraphy rotation, and had no previous exposure to the MAG3 studies in question; however, this rater was board certified in radiology with 10 years of prior clinical experience in radiology.

Order of cases was randomized per rater. For each report, the raters answered five structured questions, with an optional sixth question for additional comments:

- 1 Content of findings correct?
- 2 Impression suitable?
- 3 Requires manual correction?
- 4 Provided with this text, does it save you time in clinical routine?
- 5 Written by human or AI?
- 6 Your comments, optional:

Questions 1–4 were answered on a Likert scale (+2: strongly agree, +1: agree, 0: indifferent, -1: disagree, -2: strongly disagree). Question 5 required raters to indicate the source of the text (human, AI, or both possible).

Statistical analyses

Confusion matrices (cm) [23] were generated to compare Qwen-assigned labels to true labels, as well as raters' predictions about text sources (Question 5). The text similarity between PACS and Qwen reports was quantified by vectorising texts [24], and calculating cosine similarity [25], presented in Fig. 3. Interrater reliability (IRR) was calculated using the interrater reliability Chance-corrected Agreement Coefficients (irrCAC) python package [26]. irrCAC can provide IRR measures such as Gwet's AC1, and Fleiss' kappa, together with their respective p values. The p values allow an estimate whether the Null Hypothesis (H0) that there is no agreement beyond what would be expected purely by chance can be rejected. Both Gwet's AC1 and Fleiss' kappa take values in the range [-1, 1]. Gwet's AC1 is specifically designed for imbalanced datasets [27]. Landis & Koch [28] provided an interpretation table for numerical IRR interpretation (< 0: poor, >0: slight, >0.2: fair, >0.4: moderate, >0.6: substantial, >0.8: almost perfect). Significance between answering data from the 4 raters for Qwen/PACS reports was calculated by Wilcoxon signed-rank test [29]. Wilcoxon tests the H0

for two paired samples whether the distribution of the differences is symmetric about zero. Significance level $\alpha = 0.05$ was used

Results

MAG3 test dataset

In the present MAG3 test dataset of 50 studies (Fig. 3), 26 studies contained two regular kidneys, 8 studies included one nephrectomy-type and one regular kidney, and 16 studies included one accumulation-type and one regular kidney. On a per-kidney basis ($n=100$), 76 kidneys were regular, 8 of nephrectomy-type, and 16 of accumulation-type (Fig. 3).

Time activity curve labelling of MAG3 kidney scans

Figure 3 visualises Qwen's initial labelling performance for the dataset. On study level, Qwen correctly labelled 96.2% of the studies containing two regular kidneys, mislabelling only one out of these 26 studies. For studies involving one accumulation- or nephrectomy-type kidney, the accuracy dropped to 62.5%. On kidney level, Qwen achieved recall scores of 68.8% for accumulation, 62.5% for nephrectomy, and 96.1% for regular kidney classification. In recognition of delayed peaks for curves of regular shape, recall for delayed peak patterns was 50.0% for positive cases and 100% for negative cases.

Cosine text similarity for human and VLM reports

Cosine similarity across the entire dataset was 0.57 amongst PACS texts and slightly greater at 0.71 amongst Qwen texts. When considering only studies with one non regular kidney, cosine similarity increased to 0.58 (PACS) and 0.75 (Qwen). For studies with two regular kidneys, similarity further increased to 0.63 (PACS) and 0.78 (Qwen).

Human and VLM report quality assessment

Results from the user study split by question 1–4 are summarized in Table 1. PACS reports consistently received greater mean and mode compared to Qwen reports for questions 1 ("Content of findings correct?"), 2 ("Impression suitable?"), and 4 ("Provided with this text, does it save you time in clinical routine?"). For question 3 ("Requires manual correction?"), an inverse relationship was obtained. In the total set of all 50 MAG3 reports, PACS reports received lesser mean, and mode for question 3 compared to Qwen reports (PACS reports mean -0.74, mode -2; Qwen reports mean 0.54, mode 2). The highest overall mean rating was observed for PACS texts on question 1 ("Content of findings, correct?"; mean = 1.29).

Rating difference between text sources decreased when only Qwen reports from initially correctly labelled cases were considered. Conversely, Qwen reports from

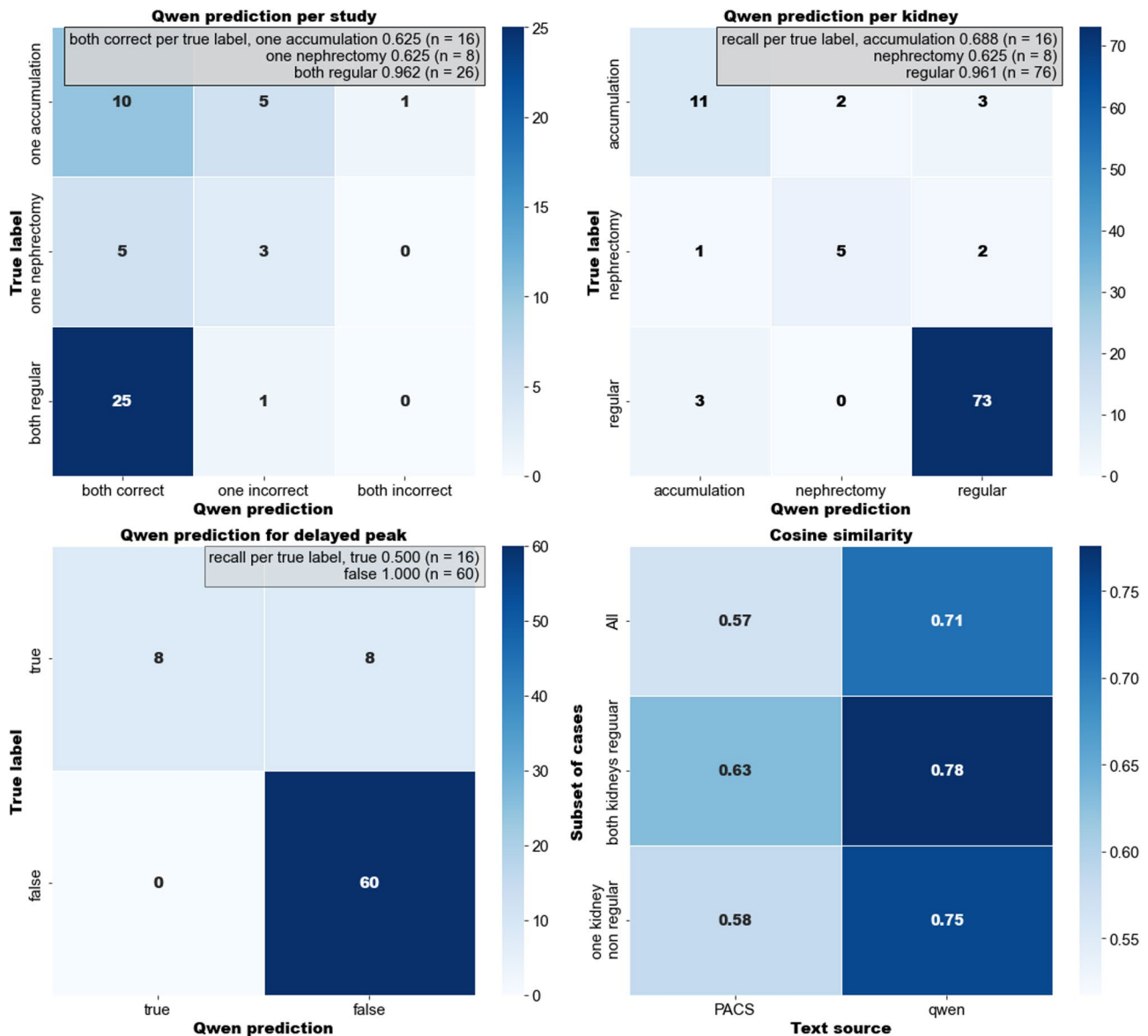


Fig. 3 Labels predicted by Qwen compared to true labels, and cosine similarity between PACS and Qwen report texts

incorrectly labelled cases received mean scores below 0 for questions 1, 2, and 4, and above 0 for question 3.

Wilcoxon Signed-Rank Test showed no significant differences for the full dataset between PACS texts, and Qwen texts for rater 1 ($p: 2.15e-01$) and for rater 2 ($p: 1.11e-01$), as well as for those cases labelled initially correctly by Qwen (rater 1 $p: 4.51e-01$, rater 2 $p: 2.15e-01$). Rater 3 reached a marginally significant difference between PACS/Qwen texts overall ($p: 4.94e-02$), but no significant difference for cases labelled correctly by Qwen ($p: 1.48e-01$). Rater 4 results were significantly different between PACS/Qwen texts overall ($p: 4.07e-04$), as well as for cases labelled correctly by Qwen (to a lesser extent $p: 2.07e-02$).

This pattern of difference in quality rating could equally be seen in Table 2 where user study results were split by rater. After inversion of values for question 3 (to reflect rater agreement), rater 1 and 2 consistently showed mean and mode scores greater than 0 for both PACS and Qwen texts. Rater 3 provided positive mean scores for PACS texts but negative mean scores for Qwen texts. Rater 4 labelled Qwen texts with the lowest scores obtained throughout the study (mode of -2).

Text source identification (PACS vs. VLM)

Figure 4 summarizes responses to question 5 ("Written by human or AI?"). All four raters together identified a proportion of 72.0% reports as definitively written either by AI or human, while for the remaining 28.0%, both

Table 1 Structured by question 1–4, likert scores from user study on answering data, comparing text quality between PACS and Qwen

Question	−2	−1	0	1	2	count	mean	std. dev.	mode
PACS reports									
1	7	9	8	70	106	200	1.29	0.99	2
2	5	8	13	78	96	200	1.26	0.93	2
3	83	45	22	37	13	200	−0.74	1.34	−2
4	6	10	18	69	97	200	1.21	1.00	2
All Qwen reports									
1	35	39	16	59	51	200	0.26	1.46	1
2	23	26	18	87	46	200	0.54	1.29	1
3	37	17	21	51	74	200	0.54	1.51	2
4	30	28	33	70	39	200	0.30	1.33	1
Qwen reports, if Qwen labelling correct									
1	15	27	12	44	46	144	0.55	1.37	2
2	9	16	11	67	41	144	0.80	1.15	1
3	34	15	13	41	41	144	0.28	1.55	1
4	11	18	24	56	35	144	0.60	1.20	1
Qwen reports, if Qwen labelling incorrect									
1	20	12	4	15	5	56	−0.48	1.43	−2
2	14	10	7	20	5	56	−0.14	1.37	1
3	3	2	8	10	33	56	1.21	1.15	2
4	19	10	9	14	4	56	−0.46	1.36	−2

Table 2 Structured by rater 1–4, likert scores from user study on answering data, comparing text quality between PACS and Qwen

Rater	−2	−1	0	1	2	count	mean	median	mode
All reports									
1	16	59	40	122	163	400	0.8925	1	2
2	60	21	55	74	190	400	0.7825	1	2
3	37	70	42	225	26	400	0.3325	1	1
4	80	58	12	74	176	400	0.52	1	2
PACS reports									
1	0	30	14	54	102	200	1.14	2	2
2	16	12	29	42	101	200	1	2	2
3	6	14	15	141	24	200	0.815	1	1
4	9	8	3	25	155	200	1.545	2	2
Qwen reports									
1	16	29	26	68	61	200	0.645	1	1
2	44	9	26	32	89	200	0.565	1	2
3	31	56	27	84	2	200	−0.15	0	1
4	71	50	9	49	21	200	−0.505	−1	−2

sources appeared possible to the readers. Among the subset of reports confidently labelled as either human or AI-generated, the raters collectively reached an accuracy of 61.5%.

A near-perfect linear relationship ($R^2 = 0.998$) was obtained when plotting mean quality scores for Qwen-generated texts (questions 1–4, values inverted for question 3) over rater accuracy in identifying text sources (question 5).

Interrater reliability

Overall interrater reliability (IRR) measured by Gwet's AC1 (0.288, fair according to Landis & Koch [28]) was marginally greater than Fleiss' kappa (0.238, fair). Both

IRR measures yielded highly significant p-values ($p < 0.05$), thus rejecting H_0 that agreement was purely chance-based. Pairwise IRR measures between raters (Fig. 4) closely resembled global values. The one rater who differed most from the remaining three was not rater 4, but rater 3.

Restricting IRR calculations to cases with an initially correct VLM assessment of the curve type slightly increased global Gwet's AC1 (to 0.291), whereas Fleiss' kappa slightly decreased (to 0.228). Thus, IRR was not substantially influenced by the correctness of Qwen labelling.

When transferring the answers from questions 1–4 (originally rated on a Likert scale of −2 to +2) into a

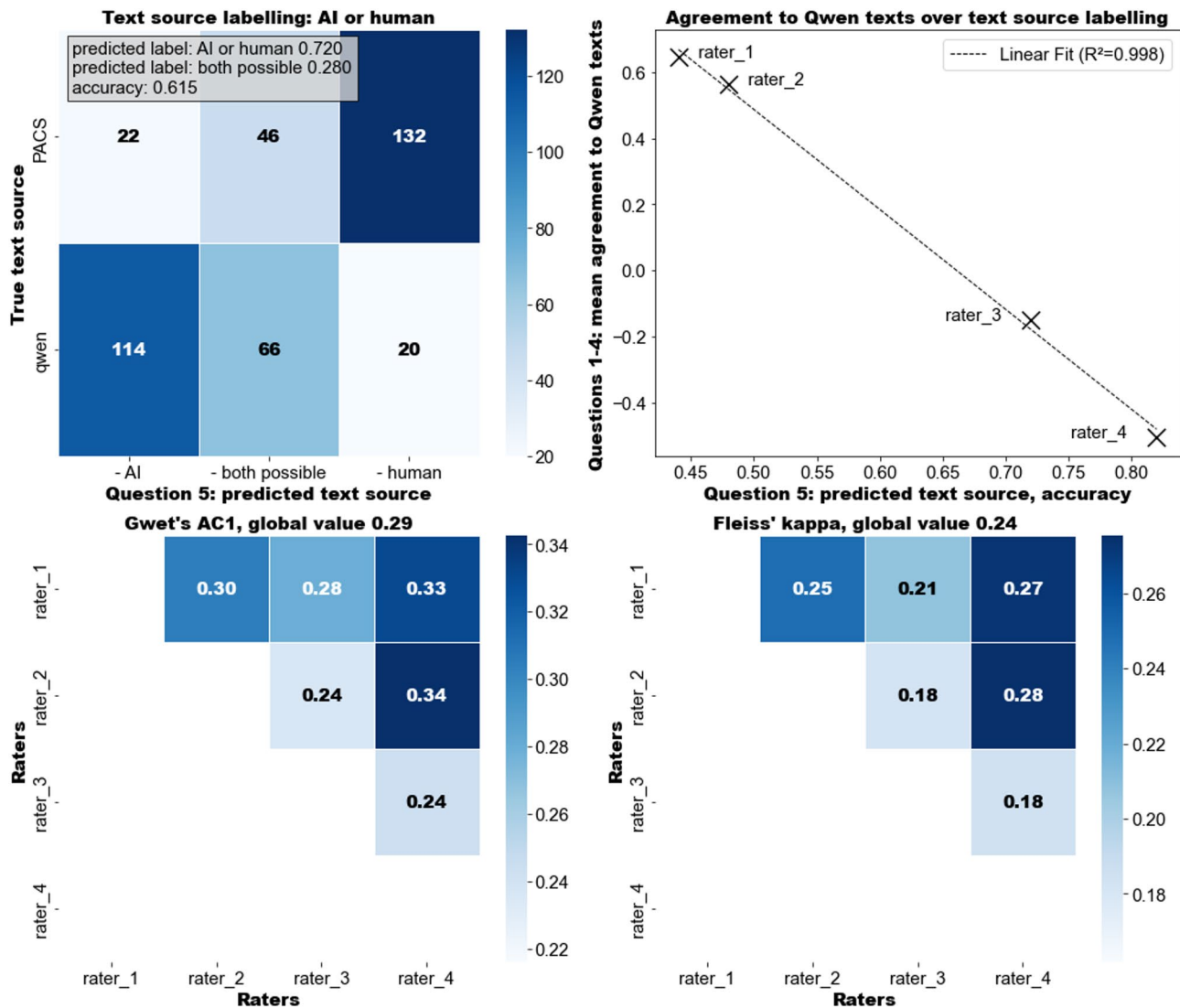


Fig. 4 Text source labelling (question 5), and IRR by Gwet's AC1 and Fleiss' kappa

simplified scale of agreement (>0), indifference ($=0$), and disagreement (<0), global Gwet's AC1 rose to a moderate level (0.527), while global Fleiss' kappa increased slightly, remaining at a fair level (0.286).

Discussion

Clinical application

In this study, the ability of a state-of-the-art VLM (Qwen 2.5-VL-72b) to generate accurate MAG3 nuclear medicine reports was evaluated. If reliable in clinical operations, such systems could potentially relieve healthcare staff from administrative burdens, enhancing hospital workflow and operational efficiency. A blinded user study involving four experienced nuclear medicine raters generally favoured human-written PACS reports over Qwen-generated reports regarding text quality (questions 1–4).

However, the perceived quality of Qwen-generated reports improved notably when limiting analysis to cases with patterns correctly labelled in the beginning by Qwen. The generalizability of these findings is limited by our single-centre dataset, small test sample size ($n=50$), and deliberate enrichment with pathological cases, suggesting that extensive validation on larger, multicentre datasets will be essential before clinical implementation.

Report text quality

Two of the four raters identified no statistically significant differences in quality between PACS and Qwen-generated texts. Conversely, the other two raters found significant or partially significant differences favouring human-generated reports. The consistent human preference observed across raters, combined with the ability to distinguish AI-generated reports from human-authored

ones (Fig. 4), highlights current limitations in VLM performance that render it insufficient for routine clinical integration. While these results provide a crucial benchmark for AI capabilities in nuclear medicine reporting, they transparently demonstrate that current VLM technology requires substantial advancement before clinical deployment can be considered.

The labelling performance of the VLM clearly was best for regular kidneys (Fig. 3). This was true on study level and on kidney level. Studies with pathological kidneys were labelled with a substantially lesser performance. This initial mislabelling lead to measurable differences in later report generation. The greater text similarity within the set of texts produced by Qwen compared to PACS text seems logical considering the highly structured text generation procedure.

Identification of AI/human author

There is much debate whether texts generated by AI are recognisable as such. In the present study, raters felt confident about labelling text source in the majority of the all blinded cases (fraction of 72.0%). Their overall accuracy (61.5%) clearly beats a random dice (Fig. 4). Among all raters, the ability to accurately identify the source of a report (AI or human-generated, question 5) very strongly correlated with their overall quality assessments (Fig. 4, $R^2=0.998$). Due to the correlational nature of this relationship, it remains uncertain whether accurate identification of the text source leads to differences in perceived quality or if quality differences facilitate accurate identification—or a combination of both factors. The assessment of VLM quality was potentially influenced by rater study biases, particularly the prior exposure of raters 1–3 to some MAG3 cases during clinical routine, which could have affected their ability to objectively evaluate report quality. Additionally, the significant inter-rater variability observed throughout our study (IRR measures indicating only fair agreement) raises important questions about the consistency of VLM quality assessment and suggests that individual rater preferences or expertise may substantially influence evaluation outcomes. Notably, rater 4, who had no prior exposure to the MAG3 studies but substantial overall clinical experience, achieved the highest accuracy (82.0%) in text-source identification; suggesting clinical experience may enhance the capability to distinguish AI-generated texts from human-authored texts, while simultaneously demonstrating that familiarity with specific cases may not be the primary factor in accurate source identification.

Interrater reliability

IRR by Gwet's AC1 was consistently greater than that by Fleiss' kappa, as expected mathematically for an imbalanced dataset; data imbalance being a typical feature of

medical datasets [27]. Greatest IRR (AC1 = 0.527, moderate) was obtained when the answering Likert scale was transformed into a pure dis/agreement scale. Overall, these findings emphasize the role of scale transformation in enhancing IRR, with a 3-point scale yielding more consistent agreement among raters.

Limitations

The current study faces several limitations that affect the generalizability and clinical applicability of our findings. The single-centre design restricts the external validity of results, as other institutions may employ different imaging protocols, patient populations, or reporting practices that could substantially affect VLM performance. Our limited test dataset of 50 cases, while adequate for initial evaluation, represents a relatively small cohort that may not capture the full spectrum of clinical presentations encountered in routine practice.

The dataset was deliberately enriched with pathological cases, creating a balanced distribution of 26 physiological to 24 pathological cases (Fig. 3). While this enrichment strategy provided more balanced data for evaluation purposes, it creates an artificial distribution that differs significantly from real-world clinical scenarios where negative findings typically dominate. In routine practice, where the majority of studies may be normal, VLM performance could differ substantially from our findings. Data imbalance is well known in medical data analysis, with plenty of examples in the literature regarding appropriate statistical analysis methods [10, 27].

Rater study biases may have influenced our quality assessments. Three of four raters had prior clinical exposure to some included MAG3 studies during routine practice, potentially affecting their objectivity in evaluating report quality. The significant inter-rater variability observed (IRR measures indicating only fair agreement) raises concerns about the consistency of VLM quality assessment and suggests that individual rater preferences substantially influence evaluation outcomes.

Only studies with two kidneys were included, excluding important clinical scenarios such as post-nephrectomy patients or kidney transplant recipients who are also included in the MAG3 patient population. Studies with only one kidney or post-transplant cases would require distinct analytical approaches tailored to their specific data patterns. Reports in this study followed a predefined template, and specific diagnostic questions from referring physicians were not incorporated into the approach, limiting the clinical relevance of generated reports.

The current VLM performance, as demonstrated by consistent human preference and distinguishable AI-generated content, is insufficient for routine clinical integration. While our results provide a crucial benchmark for AI capabilities in nuclear medicine reporting, they

clearly indicate that substantial technological advancement will be required before clinical deployment can be considered.

Conclusion

In the present work, a workflow using a pretrained VLM has been used to interpret MAG3 renal scans that are extracted from the PACS system and used for manual image analysis. The task of report generation was broken down into sub-tasks, manageable by the employed models. The approach is promising, especially with regards to the possibility of seamless integration into the manual reading of images, as no new data presentation needs to be implemented. Novel hybrid human-AI processes in MAG3 report writing would fulfil the predictions of the early AI thinkers [2]. However, the accuracy of the current approach is insufficient for use in routine clinical practice. The emergence of novel, hybrid forms of human-AI collaboration inevitably will raise ethical considerations [30], including issues such as algorithmic biases and potential deskilling of human medical experts [31]. Still, the approach is promising, and future studies are warranted.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13550-025-01323-6>.

Supplementary Material 1. Python-based cropping of clinical MAG3 image input data (.py).

Supplementary Material 2. AI-based analysis prompts for cropped MAG3 input data (.txt).

Supplementary Material 3. Python-based report generation (.py).

Supplementary Material 4. AI-based analysis prompts for cropped MAG3 input data (.py).

Supplementary Material 5. Python-Qwen flowchart (.png).

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Author contributions

WAB, CLC, FS, FAZ, NG, CMB, CM, CB, PK, KZ, KS, AR, and RS contributed to the study conception and design. Material preparation, data collection and analysis were performed by WAB, CLC, and RS. The first draft of the manuscript was written by WAB. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability

All data generated or analysed during this study are included in this published article.

Declarations

Ethics approval and consent to participate

This study was performed in line with the principles of the Declaration of Helsinki. The study received an ethics waiver (Req-2024-01348) from

the Cantonal Ethics Committee Bern (Kantonale Ethikkommission für die Forschung, Gesundheits-, Sozial- und Integrationsdirektion), dated 12 November 2024. All patients included had provided the general consent in written form.

Consent for publication

Not applicable.

Competing interests

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